Pop Stars Have Something to Teach CU Professors:
How Low Can You Go with Sentence Embeddings to Stay Useful?

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Abstract

How much embedding dimensionality is needed for a given corpus? Can a model trained on a simpler domain transfer effectively to semantically richer tasks?

To answer these questions, we study how domain and embedding dimensionality interact in unsupervised sentence encoders. Using two extreme corpora, **Taylor Swift lyrics and Nakul Verma's papers**, we train **contrastive BERT models from 4D to 767D** and evaluate geometry and transfer learning properties.

Surprisingly, lower-dimensional **models trained on lyrics (4–256D) outperform paper-trained models on STS-B**, with papers only catching up at 767D. This suggests that simple, high-frequency text can be an efficient pretraining source for compact encoders, while complex text needs far more dimensions to expose useful structure.

1 Introduction

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1.1 Related Work

Recent work on sentence embeddings has focused on understanding and improving the geometry of representation spaces. I'll briefly summarize the papers we referenced as we brainstormed and implemented our project.

Wang & Isola (2020) Wang and Isola [2020] demonstrate that contrastive learning optimizes two competing objectives—alignment of similar examples and uniformity of representations on the hypersphere—providing a geometric framework that underlies many modern embedding methods. Gao et al. (2021) Gao et al. [2021] introduce SimCSE, showing that simple dropout-based positive pairs produce state-of-the-art unsupervised sentence embeddings, with supervised variants further improving semantic similarity performance. Reimers & Gurevych (2019) Reimers and Gurevych [2019] propose Sentence-BERT, a siamese BERT architecture enabling efficient cosine-similarity–based retrieval and establishing a strong benchmark for supervised semantic embedding tasks.

Li et al. (2020) Li et al. [2020] investigate the intrinsic dimensionality of contextual embeddings, finding strong anisotropy and proposing geometric corrections to improve isotropy and downstream

38th Conference on Neural Information Processing Systems (NeurIPS 2024).

performance. Wang & Liu (2021) Wang and Liu [2021] analyze the divergence properties of InfoNCE-style losses, explaining how different contrastive objectives induce distinct geometric structures in embedding space. Chen et al. (2020) Chen et al. [2020] introduce SimCLR, a vision-based contrastive framework whose insights—especially the role of augmentations and projection heads—directly inspired modern sentence-level contrastive methods. Amit et al. (2021) Amit et al. [2021] tie generalization performance to intrinsic dimensionality, suggesting that models adapt their effective dimensionality to task complexity. Finally, Hu et al. (2023) Hu et al. [2022] present LoRA, showing that learned low-rank subspaces can efficiently adapt large language models—reinforcing the broader theme that meaningful linguistic structure often lies in low-dimensional manifolds.

2 Methodology

ALENA (except for data preprocessing)

2.1 Base model

`BERT-base-uncased` with mean pooling. A projection head maps pooled representations to a target dimension $d \in \{4, 16, 64, 256, 767\}$ followed by L_2 normalization.

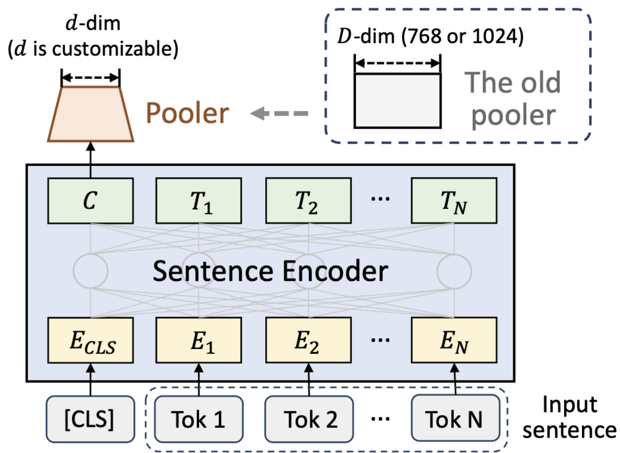

Figure 1: BERT+Pooler architecture. Image credit: https://arxiv.org/pdf/2310.15285.

2.2 Training domains

Each song and each paper section is treated as one document, and neighboring sentences within a document form positive pairs. The corpora are token-matched for training compute parity:

1) Taylor Swift lyrics (short, repetitive, low-entropy)

2) Nakul Verma's ML papers (long, information-dense)

2.3 Data preprocessing

Downloading, cleaning, and formatting the datasets into sentence-level documents was a multistep process.

For the lyrics dataset, there was a kaggle repository containing the entire Taylor Swift discography, which we used for our lyrics corpus. The cleanup included removing sequential duplicate sentences, filtering out song and album metadata, and splitting the lyrics into individual sentences using the NLTK sentence tokenizer.

For the papers dataset, we used the arXiv API to download Nakul Verma's papers in pure latex form. There was substantial cleaning and processing work involved from that point. First, we removed latex commands, equations, figures, tables, and references using regex patterns. Next, we split the cleaned text into sentences. These two actions required a fair amount of user judgement, and dimished the size of the usable corpus.

2.4 Training objective

We train unsupervised SimCSE-style contrastive models with multiple positive examples, sampling local neighbors within each document. Following Wang et al., 2023 Wang et al. [2023], we use a two-step training scheme:

Step 1. Freeze pooler, train encoder (3 epochs).

Step 2. Freeze encoder, train pooler (3 epochs).

2.5 Evaluations

- Geometry
  - Alignment and uniformity losses (Wang et al., 2020, Wang and Isola [2020]) on in-domain and out-of-domain test sets. Alignment measures how close positive pairs are, and uniformity captures how evenly embeddings are spread on the hypersphere.
  - t-SNE embeddings of sentences from songs and papers.
- Transfer learning: STS-B Spearman rank correlations of sentence pair embeddings.

PCA projections of the frozen BERT encoder at the same target dimensions are used as a baseline.

3 Results and Analysis

3.1 Spearman Rank on STS-B

The results on lyrics-trained models rise sharply between 4D and 256D, peaking long before the full BERT size of 768D (see Fig. 2). Paper-trained models, in contrast, only come to full potential at 767D, hinting at a much **higher intrinsic dimensionality of scientific text**. Since STS-B is out-of-distribution for both training corpora, the gap between the performance of songs- and papers-trained models cannot be attributed to lyrics being much "closer" to STS-B than papers. Instead, it reflects how contrastive fine-tuning under a low-dimensional bottleneck **reallocates capacity of the model**.

On the lyrics corpus, the contrastive task is relatively simple and repetitive, so the model can solve it without drastically reshaping the pre-trained BERT geometry. A small projection head is enough to compress lyrics-specific structure while preserving generic semantic directions that still transfer to STS-B.

On the papers corpus, the contrastive task is much harder: the model must separate dense, technical sentences within and across sections. In low dimensions, the encoder and head are forced to specialize for fine-grained distinctions, which distorts general-purpose semantics and hurts STS-B transfer. Only when we increase the projection size up to 256-767D does the paper-trained model regain enough capacity to encode both in-domain nuance and broad semantic structure, at which point it overtakes the lyrics-trained model.

The PCA baseline stays near zero Spearman rank across all projection sizes, suggesting that **linear compression alone cannot expose semantic structure** – the pooler must learn it.

3.2 Uniformity and Alignment

All plots 3 show the **alignment vs uniformity tradeoff**: higher dimensions improve uniformity but degrade alignment. For PCA specifically, uniformity losses are much smaller than alignment losses for all dimensions, indicating that embedded sentences occupy the full hypersphere but do not preserve local similarity structure.

For contrastive training, **in-domain data** (eg. songs embedded with a model trained on songs) **becomes well-spread** (good uniformity) but not overly tight, and **out-of-domain data** often collapses, achieving **artificially low alignment with poor global geometry**. This collapse likely occurs because the model is forced to project unfamiliar sentence structure into a space shaped only by in-domain

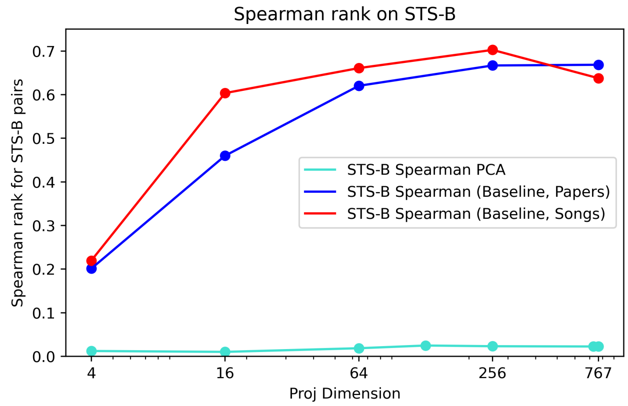

Figure 2: Spearman rank correlations for BERT+pooler and BERT+PCA variants.

positives, over-compressing local neighborhoods without preserving meaningful global structure. In addition, papers-trained models behave more similarly in terms of both losses on different domains, while songs-trained models have larger oscillations when domains are switched, hinting at more stable geometry learned by papers-based embeddings across domains.

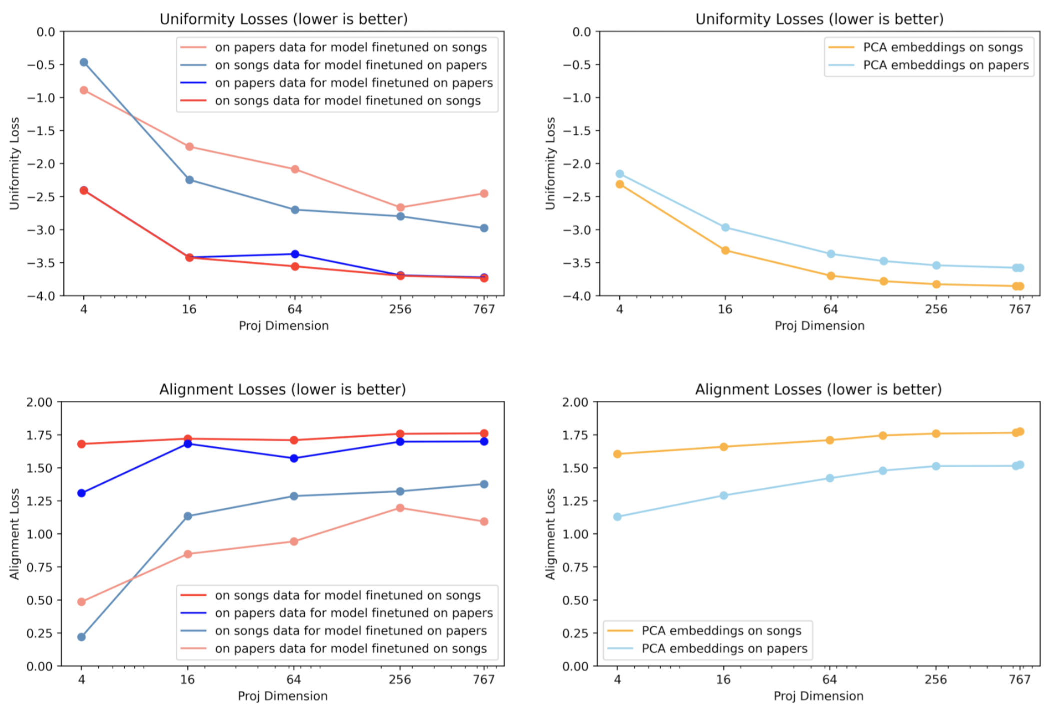

Figure 3: Uniformity and alignment losses.

3.3 Songs and Papers Clusters Embedded with t-SNE

Even with only 4 dimensions, the embeddings already separate lyrics from papers as two distinct clusters (see Fig. 4), indicating that coarse domain identity survives aggressive compression. However, in 4D the clusters are elongated and less compact, with visibly distorted shapes. At 16 dimensions, the clusters become rounder, and local neighborhoods look more stable. This suggests that increasing dimensionality from 4D to 16D does not create domain separation from scratch, but instead improves the within-domain geometry: it reduces compression artifacts and yields a smoother manifold, which is consistent with the concurrent improvement in STS-B performance.

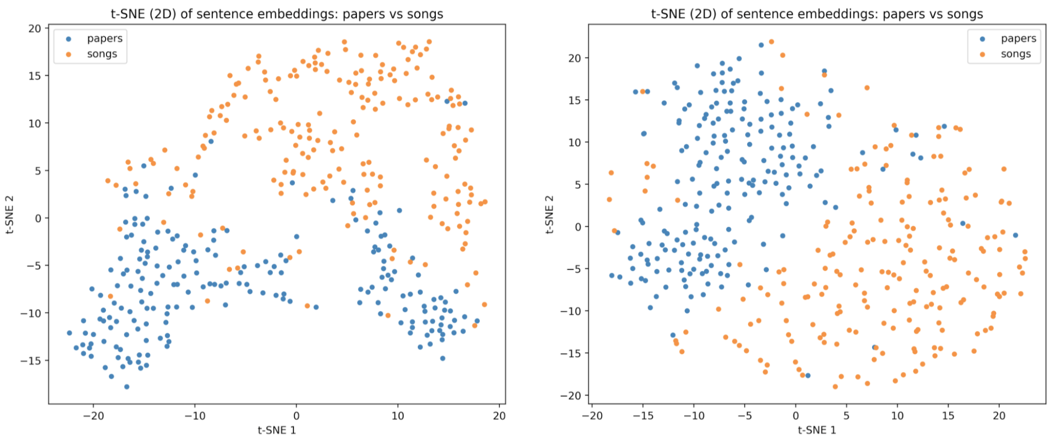

Figure 4: t-SNE results for 4D and 16D BERT+pooler models trained on songs.

3.4 Ablation on the Training Procedure

For songs-trained models, with smaller dimensions (4, 16), the two-step training process yields the best Spearman rank correlations, showing how **important the pooler is for extreme dimensionality reduction** (see Fig. 5). However, at certain higher dimensions training only the encoder yields the best results. Training only the pooler is not sufficient for learning useful representations for songs-based models.

For papers-based models, we see a similar trend, but the dimension cutoff at which the pooler stops giving much benefit is higher, somewhere between 128D and 256D (again hinting at the higher intrinsic dimensionality of scientific data). In addition, only training the pooler with the frozen BERT backbone yields much better Spearman rank results for papers-trained models than for songs-trained models – perhaps hinting to us that the frozen BERT encoder is initially better suited for academic text.

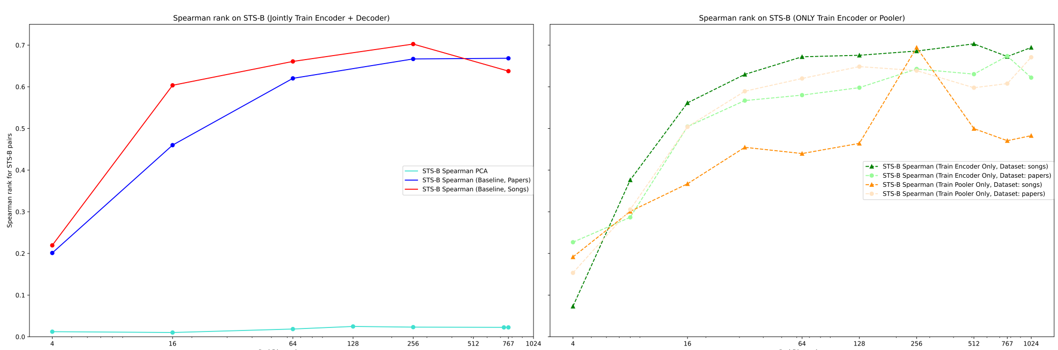

Figure 5: Ablation study on the two-step training procedure.

3.5 Challenges and Limitations

We note that all reported results depend on a specific choice of backbone (BERT-base), corpus size, and contrastive neighborhood definition. In addition, while STS-B Spearman rank results provide an insightful semantic probe, it represents only one notion of transfer learning performance.

Evaluation Criteria Selection of evaluation metrics (alignment, uniformity, STS-B Spearman rank) was an area of much thought and discussion. Drawing inspiration from Gao et al. Gao et al.

[2021], we explored enriching our unlabeled dataset with self-supervised positive pairs. We landed on sentence proximity as a straightforward and effective method to generate these pairs without additional labeling.

Contrasting Loss Finetuning By design, the constrastive loss finetuning only seeks to optimize local structure and better distribute embeddings along the unit hypersphere. However, the inherent upper bound on this optimization objective comes from the pre-trained representations provided by BERT. Our procedure is not designed to, nor is it likely to, drastically alter the global structure of the embedding space. Hence, this route will not likely yield a dramatic improvement in sentence embedding, or revolutionize the field.

Scope Due to time and resource constraints, we limited out experiments to two esoteric domains of the english language. A more comprehensive study would include a much broader array of domains, self-labeling algorithms, and backbone architectures.

BERT As our backbone, BERT-base-uncased is a well-established model for sentence embeddings. However, there are more recent encoder architectures (e.g., RoBERTa, DeBERTa, etc.) available for exploration. In addition, the past five years have seen a preponderance of investment into closed-source LLMs (e.g., GPT, Gemini, Grok, Claude, etc.) that may offer superior embedding capabilities (with the caveat that it's uncertain that cleanly demarcated encoder-decoder patterns exist in said closed models).

3.6 Future Directions

For future work, we would like to explore **additional domains** (such as news, code comments etc.), try other **embedding models beyond BERT**, use **alternative training objectives** (for instance, non-contrastive methods), enrich the suite of **evaluation methods**, and implement competing self-supervised **sentence pair labeling algorithms**.

Expanding the scope a bit, it would be interesting to move beyond all-purpose sentence embeddings and explore **task-specific embeddings** and **multimodal embeddings**, modeling language in a **hierarchical manner**, moving beyond euclidean spaces into **non-euclidean geometries**, and exploring the notion of **intrinsic dimensionality**.

4 Conclusion

4.1 Summary of Findings

- **Lyrics-trained models dominate at low dimensions (4–256D) on STS-B**, while papers-trained models only overtake at 768D $\Rightarrow$ **scientific text appears inherently higher-dimensional**.
- **Uniformity and alignment conflict** with each other, with **papers-trained models being more stable** when transitioning from in-distribution to out-of-distribution data.
- **Low-dimensional embeddings** are able to **separate domains (even at 4D)**, but more meaningful local structure emerges only at 16D or more.
- Linear compression **(PCA) is insufficient for transfer learning** tasks in our experiments: contrastive learning is necessary to expose semantic structure even at small dimensions.

4.2 Contributions of each team member

- **A.C.**: BERT+pooler architecture implementation, BERT+PCA architecture implementation and evaluation on 4-767D, partial report write-up.
- **A.K-W.**: Data collection and preprocessing for both datasets, ablation studies on the BERT+pooler architecture for both datasets with training and evaluation on 4-767D, partial report write-up.
- **M.G.**: Initial literature review, evaluation suite and graphics implementation, baseline BERT+pooler training and evaluation on 4-767D for both datasets, poster preparation, partial report write-up.

Acknowledgments

We are grateful to Prof. Nakul Verma, and our teaching assistants, Noah Bergam and Szymon Snoeck, for their feedback on our ideas.

Code Availability

All our code can be found at https://github.com/AlenaResiko/UML.

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Appendix